

Decoding the Game:

A Game-Theoretic and NLP Framework for Detecting Information Asymmetry in Financial Markets

PROBLEM STATEMENT

Can NLP-driven analysis of earnings call transcripts, grounded in game-theoretic signaling frameworks, detect private information signals embedded in managerial language — and does subsequent stock market behavior conform to the Nash Equilibrium outcomes predicted by asymmetric information models?

ABSTRACT

Financial markets are fundamentally arenas of strategic interaction under incomplete information. Executives who conduct quarterly earnings calls possess non-public knowledge about their firm's true operational health, growth trajectory, and emerging risks — knowledge that retail and even institutional investors do not directly observe. This structural gap, known as *information asymmetry*, creates a strategic environment in which the language choices of management function not merely as communication, but as signals in the formal game-theoretic sense.

This project proposes a framework that models earnings calls as a *signaling game* — drawing from the foundational work of Spence (1973) and the market microstructure insights of Kyle (1985) — wherein management constitutes the informed sender and market participants act as the receiver. Under this formulation, linguistic choices in earnings calls (specificity of forward guidance, prevalence of hedging language, directness of responses to analyst questioning) are treated as strategic signals whose informational content can be decoded through Natural Language Processing. The equilibrium concept adopted is the Bayesian Nash Equilibrium, wherein each player's strategy is optimal given their beliefs about the private types of other players.

Operationally, this study will construct a corpus of earnings call transcripts drawn from S&P 500 companies across multiple sectors, paired with post-announcement stock price movements and subsequent quarterly performance outcomes. Linguistic features will be extracted using the Loughran-McDonald financial sentiment lexicon and contextual embeddings generated via FinBERT — a domain-adapted variant of the BERT language model pre-trained on financial text. A classification model trained on these features will predict whether the embedded signal is consistent with a firm in strong or deteriorating condition. Critically, the study will evaluate whether observed market responses to these linguistic signals align with the equilibrium predictions derived from the signaling game — specifically, testing for the presence of a

separating equilibrium (where distinct firm types produce distinguishable language patterns) or a *pooling equilibrium* (where strategic mimicry renders signals uninformative).

The anticipated contribution of this work is threefold: a theoretically grounded game-theoretic model of managerial disclosure behavior; an empirically validated NLP pipeline for extracting strategic signals from financial text; and a quantitative assessment of whether equilibrium theory accurately describes real-world market responses to informational signals. The findings are expected to carry implications for market efficiency research, algorithmic trading strategy, and regulatory policy around disclosure quality.

KEYWORDS Game Theory · Signaling Games · Information Asymmetry · Natural Language Processing · FinBERT · Market Microstructure · Nash Equilibrium · Earnings Call Analysis · Financial Markets
